

Symmetry

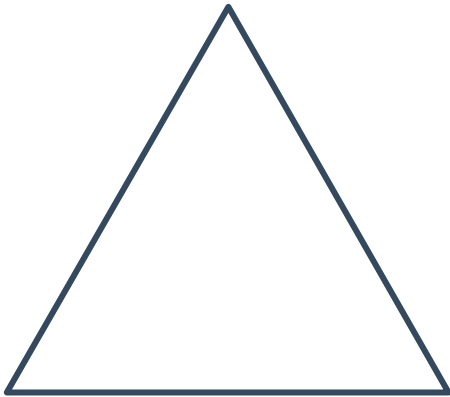


- 1 Consider all the single digit numbers shown below.
Which of these numbers have line symmetry?

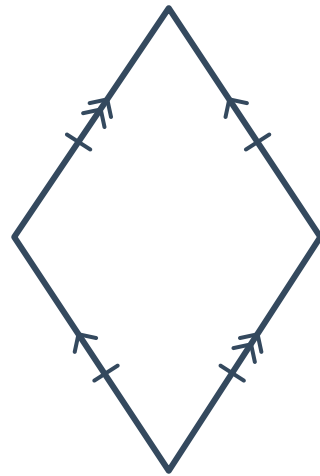
1 2 3 4 5 6 7 8 9 0

- 2 How many lines of symmetry do the following shapes have?

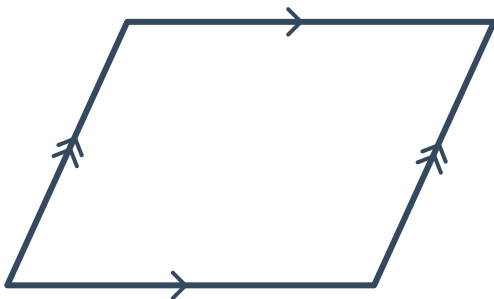
a



b



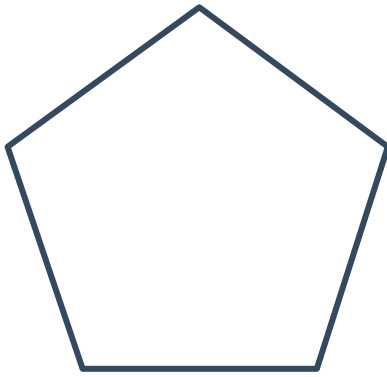
c



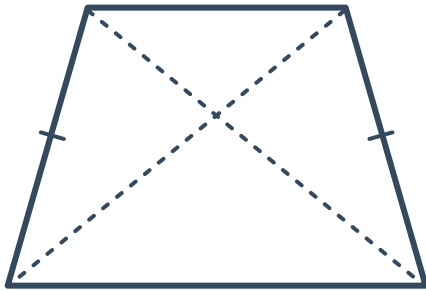
d



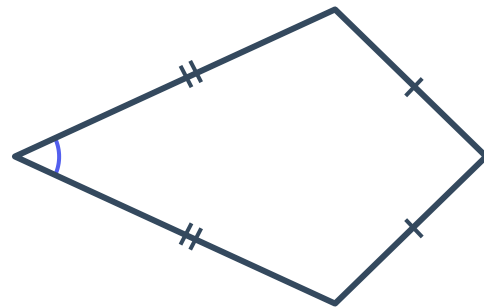
e



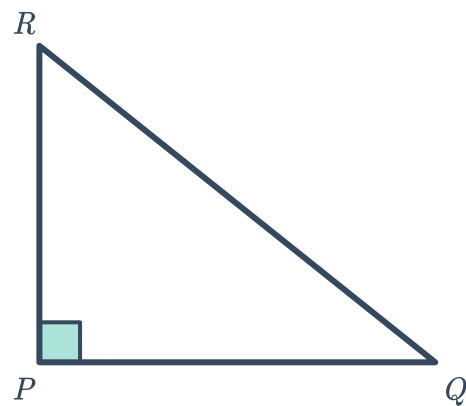
g



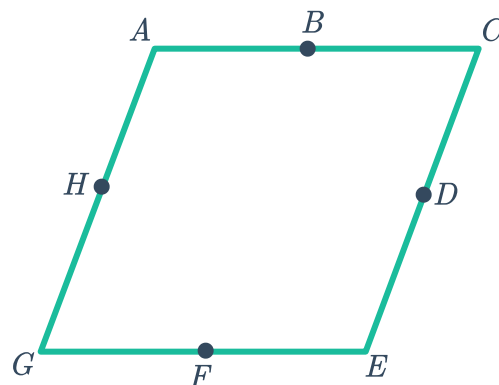
h



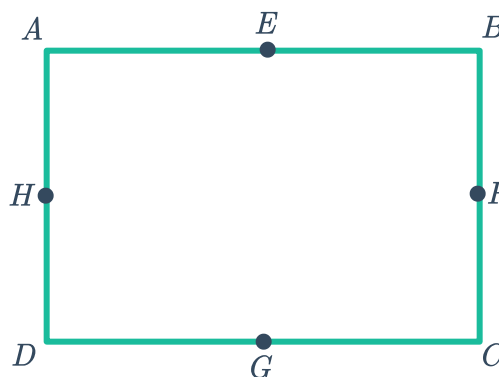
- 3 Consider the $\triangle RPQ$ shown. The triangle is reflected across $\overline{PQ}$. Identify the shape formed by the triangle and its reflection.



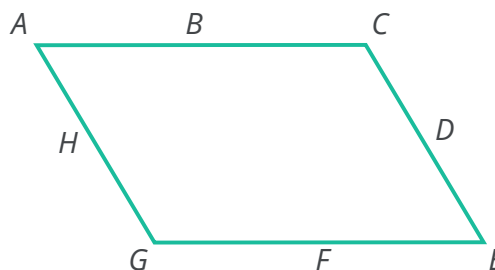
- 4 Consider the following rhombus.
The points B, D, F and H represent the midpoints of the sides of the rhombus.
Write down the pairs of points that can be joined to form a line of symmetry for the rhombus.



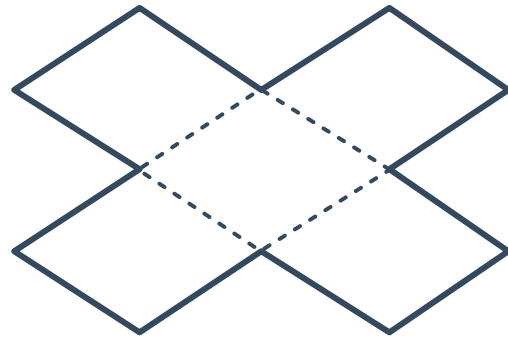
- 5 Consider the following rectangle.
The points F, G, H and E represent the midpoints of the sides of the rectangle.
Write down the pairs of points that can be joined to form a line of symmetry for the rectangle.



- 6 Consider the following parallelogram.
The points B, F, H and d represent the midpoints of the sides of the parallelogram.
Write down the pairs of points that can be joined to form a line of symmetry for the parallelogram.



- 7 The following shape is constructed from five identical rhombuses. How many lines of symmetry does this shape have?



- 8 Determine whether the following diagrams have rotational symmetry.

a



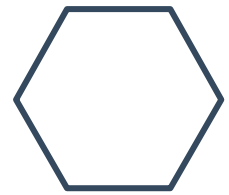
b



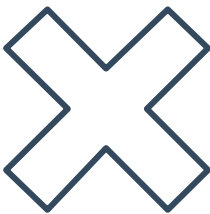
c



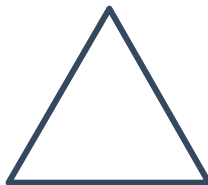
d



e



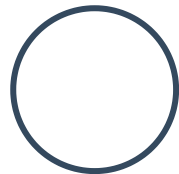
f



g



h



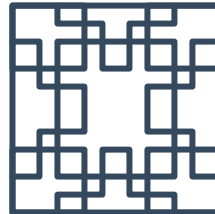
i



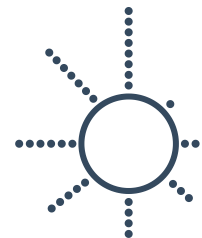
j



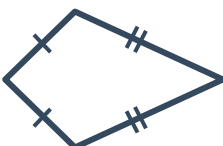
k



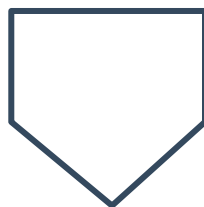
l



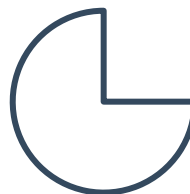
m



n



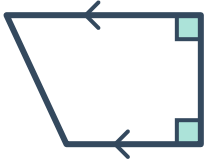
o



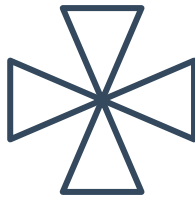
p



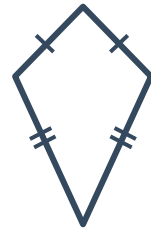
q



r



t



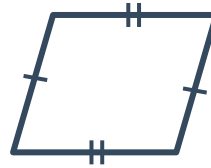
u



v



w

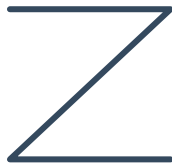


9 Determine whether the following letters have rotational symmetry.

a



b



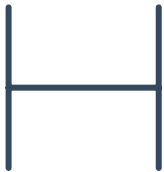
c



d



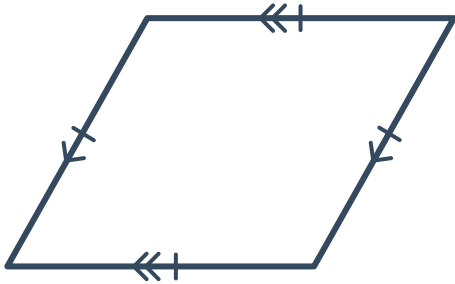
e



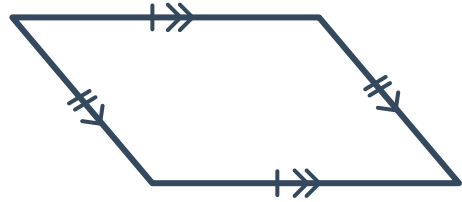
10 State the order of rotational symmetry for the following shapes.



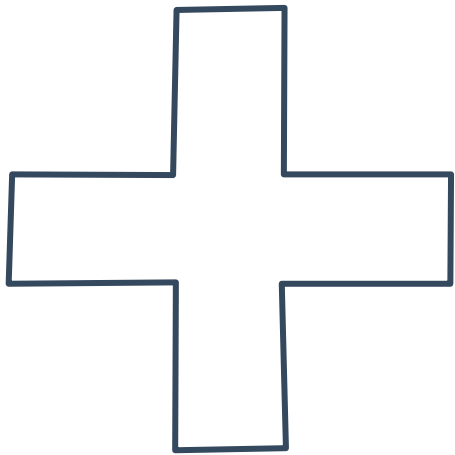
a



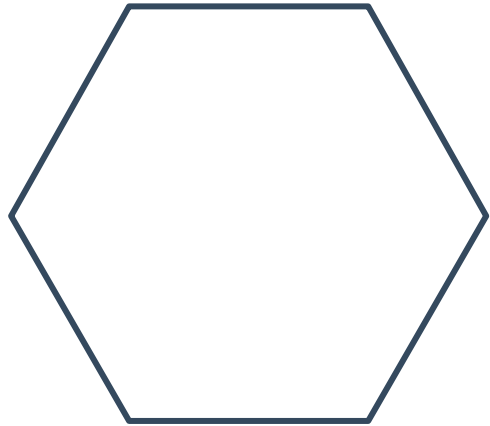
b



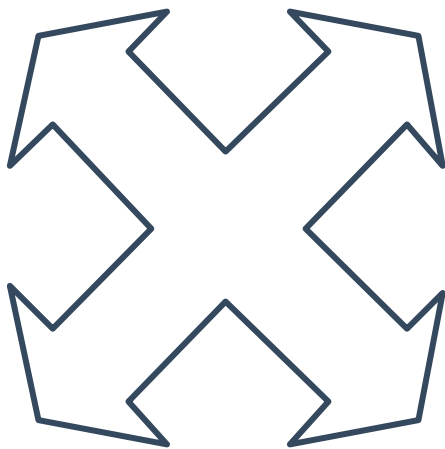
c



d



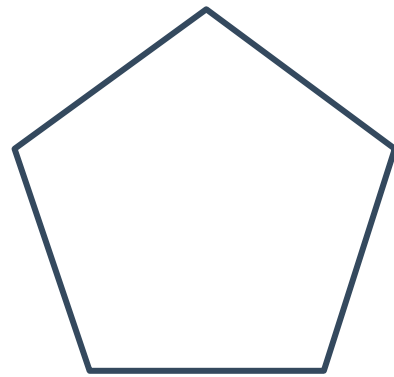
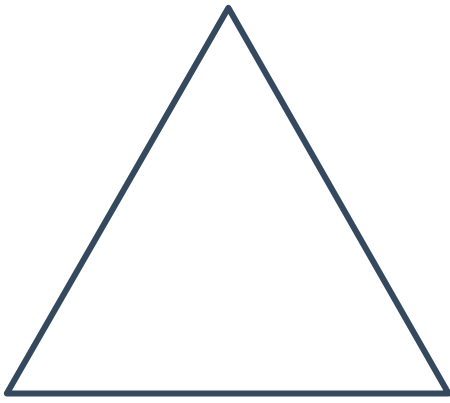
e



f



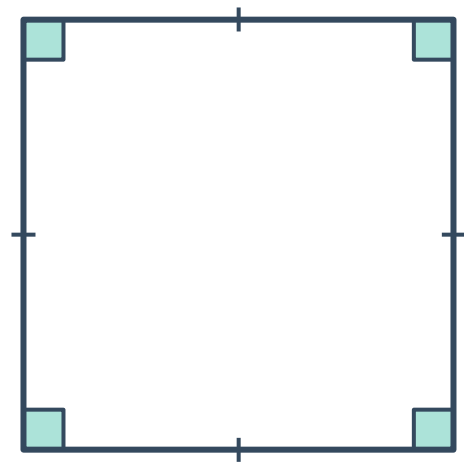
g



i



j

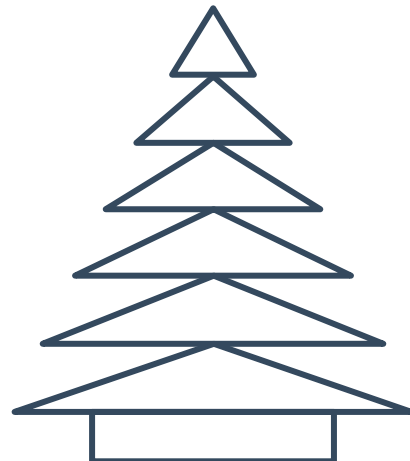


11 State the type(s) of symmetry each of the following diagrams have.

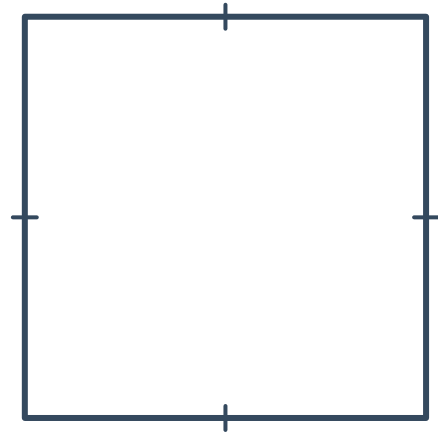
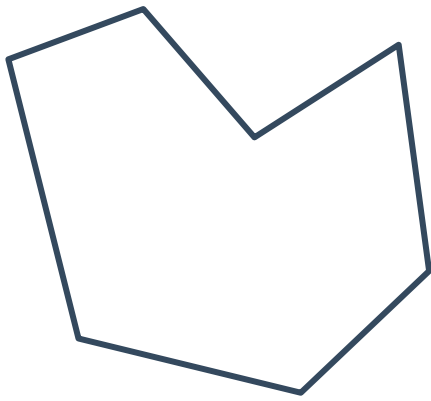
a



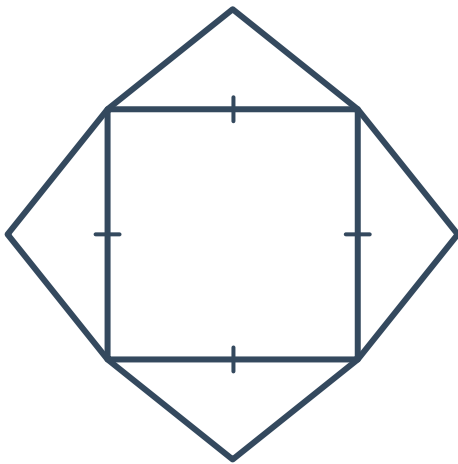
b



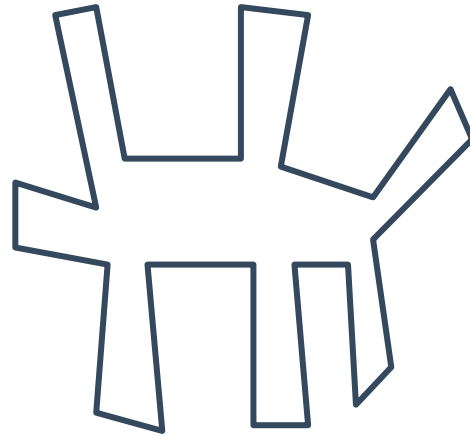
c



e



f

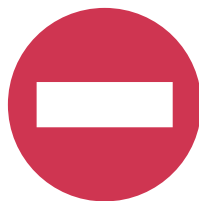


12 If a shape has a rotational symmetry of order 2, state the size of each angle of rotation.

13 Do all isosceles triangles have rotational symmetry?

14 Do the following road signs have rotational symmetry?

a



b



15 Consider the road sign shown.

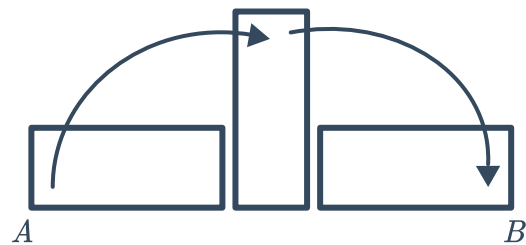


- a State the order of rotational symmetry.
- b Does the road sign have line symmetry?



16 Dave wants to get a heavy rectangular box from point A to point B by repeatedly turning it over. The width of the box is 30 cm and its length is twice its width.

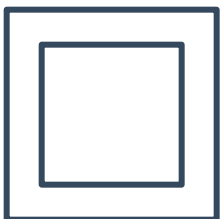
- a The diagram shows the distance travelled after the first and second turn. After 3 turns, what distance along the ground has been covered starting from point A ?
- b If the distance from point A to point B is 1620 cm, how many times must the box be turned?



Additional questions

17 Determine whether the following shapes have rotational symmetry:

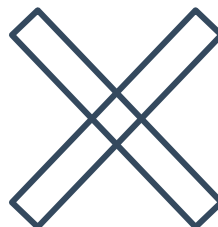
a



b



c



d



18 If a shape has a rotational symmetry of order 4, state the size of each angle of rotation.

19 Do all equilateral triangles have rotational symmetry?

